

Blind Faith

This month I'd like to talk about science: the great fiery crucible in which the engines of invention are forged; the true birthplace of technology and all that is good; the real reason why you are about to read this magazine under the bright light of an incandescent bulb, amongst other "miracles".

The reason I want to talk about science is thanks to a rather silly video that's been doing the rounds on Facebook lately. It's an old video by a well known evangelist who goes by the name of Ray Comfort. Like all bad ideas, the video keeps resurfacing from time to time and is an attempt to disparage science and scientific thought. The video features Ray going around a US campus (presumably UCLA) interviewing students and professors alike, asking them if they "believe" in evolution. The video adds an unhelpful and cherry picked definition of evolution too at the beginning, which is attributed to LiveScience: Evolution can turn dinosaurs into birds, apes into humans, and amphibious mammals into whales.

The interviewees – almost all of them biology majors, PhDs, physicists and biochemists – are then cornered into presenting "observable evidence that can stand as proof of evolution". This is where the video gets interesting. The respondents to this sneak attack, experience a visible deer-in-the-headlights moment. Eventually, when they recover and talk of the fossil record and evolution over "millions of years" he quickly jumps at the time period stating "so that means it can't be observed".

When the respondents talk of the beaks of Galapagos finches changing to suit conditions of specific islands they evolved on, our friend asks the interviewees to give evidence of "change of kind" not minor adaptations. Ray presumably wants photographic evidence of the precise moment when say a slug like creature developed a vertebra or when a primordial vertebrate became a dog. Maybe even photographic evidence won't do. Maybe he wants to see it happen in front of his eyes. That's like observing a tree for a nanosecond, not perceiving any growth and saying trees don't age at all. While

everyone knows that some trees like the mighty sequoias grow to great heights over a thousand years in some cases.

A few brilliant tricks of editing later when his respondents are seen dumbfounded (who wouldn't be?) finally telling him that they can't present "observable" evidence of the kind he expects, he asks his victims another question: "Do you know what belief in something without evidence is known as?" No Mr. Guy With Camera, what is it known as? "Blind faith," he says emphatically. You can watch the video here if you wish: <http://dgit.in/BlndFth>

Now as I sit here in the comfort of a chair I can say things like "if I was being interviewed I'd show him". But one look at Indian news channels and we all know what the effects of being in front of a camera, badgered by a confident inquisitor can do to the best of orators. I'm no orator but I can certainly try to break down his argument piece by piece on paper. My first response is that this person should really try to not discount the effects of time. We tend to think in terms of our own measly lifetimes but the processes that shape things around us are far far far more lengthy. It's hubris to give our own life spans so much importance. The second has to do with the fossil record. Undoubtedly, even if I were to take this gentleman to a museum of natural history, he'd want rocks to literally depict "change of kind" not minor adaptations. These breakages in the record are what are known as missing links. Science may not have all the answers, but it's at least trying to search for them.

Besides, any person with even a remote interest in dinosaurs would be able to tell you that the creation of a fossil needs a very special set of conditions to be fulfilled. It's so rare an occurrence that we're lucky to even have as many fossils as we do today.

So yes the fossil record is evidence. Genetics is evidence. It's too bad it doesn't constitute as evidence to you Mr. Ray, but it doesn't discount science or scientific thought in any way. If we insist on only personally observable evidence to constitute as proof of something only astronauts should "believe" that the earth is round



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"If we insist on only personally observable evidence to constitute as proof of something, only astronauts should *believe* that the earth is round"

right? No one else on earth has actually seen evidence for it being round.

I'd like to conclude by saying that there is something that lies between the two extremes of blind faith on one side and personally observable evidence on the other – and that is logic. If an idea is logical, by extrapolation you can arrive at astounding results. The concept of gravitational waves was a logical idea, Mendeleev's periodic table predicted the existence of elements that weren't discovered yet. Evolution, which has been called "the single best idea that anyone has ever had" is exactly such a concept. 

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